

ATPL EASA STUDY GUIDE | 033 - FLIGHT PLANNING

Chapter 4: ICAO ATS Flight Plan (FPL) Form
Item-by-Item

Syllabus Source: CAE Oxford (pp. 121-164) | Pass Target: >= 90% AviationExam

1. THE ICAO MODEL FLIGHT PLAN FORM STRUCTURE

The standard ICAO Flight Plan Form is used globally for Air Traffic Control coordination. Every item must be encoded precisely:

Item Number & Title	Regulatory Code Formats & Allowed Entries	Operational Significance & Rules
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Item 7: Aircraft ID	Max 7 alphanumeric characters (e.g. IBE3412 or EC-MZZ).	Exact callsign matching radio communication.
Item 8: Flight Rules & Type	Rules: I (IFR), V (VFR), Y (IFR changing to VFR), Z (VFR changing to IFR). Type: S (Scheduled), N (Non-scheduled), G (General), M (Military), X (Other).	Y and Z flights require specifying the changeover point in Item 15!
Item 9: Type & Wake Cat	Number + 4-letter ICAO type (e.g. A320, B738). Wake turbulence: L (<7,000 kg), M (7,000 to 136,000 kg), H (>=136,000 kg), J (Super A380).	Dictates ATC wake separation intervals!
Item 10: Equipment & Cap	10a COM/NAV (S for standard VHF/VOR/ILS, D, G for GNSS, R for PBN approved). 10b SSR (Mode S, ADS-B).	R requires PBN/ specification in Item 18!
Item 13: DEP & EOBT	4-letter ICAO departure aerodrome (e.g. LEMD) + 4-digit Estimated Off-Block Time (UTC).	If no ICAO code, enter ZZZZ and specify in Item 18 DEP/.
Item 15: Speed, Level & Route	Speed: N (knots, 4 digits, e.g. N0450) or M (Mach, 3 digits, e.g. M078). Level: F (Flight Level, 3 digits, e.g. F350) or A (Altitude in 100s ft). Route: ATS airways and waypoints.	Speed must be initial true airspeed (TAS).
Item 16: DEST, EET & Alternates	4-letter destination (e.g. LFPG) + Total EET (hours:min, e.g. 0145) + 4-letter Alternate 1 & 2.	Total EET is time from take-off to overhead destination.
Item 18: Other Information	PBN/ (e.g. B1D1O1S2), DOF/ (Date of Flight YYMMDD), REG/ (Registration), EET/ (FIR boundary estimates), RVR/.	0 (zero) indicates no special information.
Item 19: Supplementary Info	E/ (Endurance hhmm), P/ (Persons on board), R/ (Emergency radio), S/ (Survival equipment), J/ (Jackets), D/ (Dinghy).	Transmitted only for Search and Rescue (SAR).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): ICAO Flight Plan Form Item 8 Rule 1

Rule 1 (Yankee 'Y'): Flight departs under IFR in controlled airspace, then cancels IFR enroute to proceed VFR to destination. Rule 2 (Zulu 'Z'): Flight departs under VFR from uncontrolled field, then files/picks up IFR clearance in flight to join airways. Crucial Form Rule: In Item 15, specify the exact navigational fix where the change occurs, followed by space and the new rule (e.g. 'MERIT VFR' or 'TOU IFR').

2. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)**[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Filling an ICAO ATS Flight Plan**

SCENARIO (Standard EASA FPL Encoding Drill): A scheduled commercial flight is operated by Iberia: - Callsign: IBE6501, Aircraft: Airbus A350-900 (MTOM = 280,000 kg) - Flight Rules: Entirely IFR - Initial Cruising Speed: TAS 480 knots; Cruising Level: FL 390 - Departure: Madrid Barajas (LEMD) at 14:30 UTC - Destination: New York JFK (KJFK); Total Estimated En-Route Time: 7 hours 45 minutes - Alternate: Boston Logan (KBOS) - Usable Fuel Endurance on board: 10 hours 30 minutes; Total Persons on Board: 312 QUESTION: How are Items 7, 8, 9, 13, 15, 16, and 19 encoded? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Item 7 (Aircraft Identification) = IBE6501 Step 2: Item 8 (Flight Rules & Type): - Flight Rules: I (IFR) - Type of Flight: S (Scheduled air service) - Entry: I S Step 3: Item 9 (Number, Type & Wake Turbulence): - Type: A359 - Wake Turbulence Category: MTOM = 280,000 kg (> 136,000 kg) -> Category H (HEAVY)! - Entry: A359 / H Step 4: Item 13 (Departure Aerodrome & Time): - LEMD 1430 Step 5: Item 15 (Speed & Level): - Speed: TAS 480 kt -> N0480 (N followed by 4 digits) - Level: FL 390 -> F390 (F followed by 3 digits) - Entry: N0480 F390 [Route...] Step 6: Item 16 (Destination, Total EET & Alternate): - KJFK 0745 KBOS Step 7: Item 19 (Endurance & POB): - E/ 1030 P/ 0312 FINAL ANSWER: Encoded exactly as above! Zero errors, fully compliant with ICAO Doc 4444.

[!] EXAM TRAP (AVIATIONEXAM) - The FPL Speed & Wake Turbulence Traps

- Speed Format: 'N' (Knots) MUST have 4 digits (e.g. 120 kt = N0120; 450 kt = N0450)! - Wake Turbulence: Heavy (H) is $\geq$ 136,000 kg (136 tonnes). Medium (M) is 7,000 to 136,000 kg. Super (J) is Airbus A380 only! - Flight Rules Y vs Z: Y = IFR changing to VFR. Z = VFR changing to IFR (Remember: Alphabetical order V->Z!).

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: On an ICAO Flight Plan Form, what does flight rule code 'Y' indicate?**

- [A]: Pure VFR flight
- [B]: A flight planned to operate initially under IFR, followed by one or more subsequent changes of flight rules to VFR
- [C]: A flight in yellow airspace
- [D]: Military flight

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Code Y indicates an IFR flight that changes to VFR en-route. Code Z indicates a VFR flight that changes to IFR en-route.

Q2: In Item 9 of the ICAO Flight Plan, what is the Wake Turbulence Category for an aircraft with a Maximum Ce

- [A]: M (Medium)
- [B]: H (HEAVY), because MTOM is greater than or equal to 136,000 kg
- [C]: J (Super)
- [D]: L (Light)

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Heavy (H) wake turbulence category applies to all aircraft with an MTOM of 136,000 kg (300,000 lb) or more.

Q3: How must an initial true airspeed of 210 knots be entered into Item 15 of an ICAO Flight Plan?

- [A]: 210K
- [B]: N0210 (the letter N followed by exactly four digits)
- [C]: TAS210
- [D]: KT0210

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Airspeed in knots must be expressed by the letter N followed by 4 digits, padding with leading zeros if necessary (e.g. N0210).